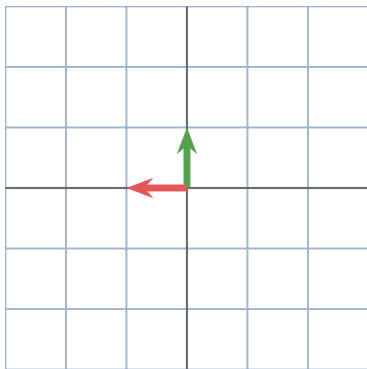
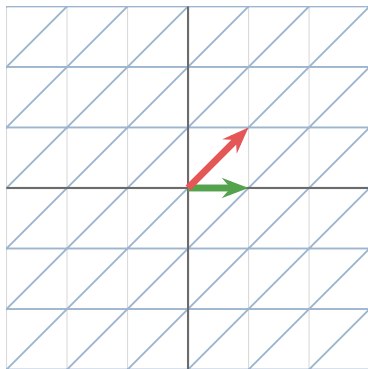


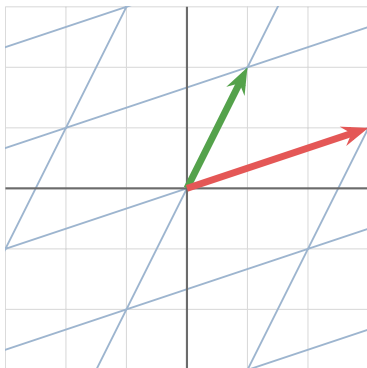
Rotation by 90° : $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$



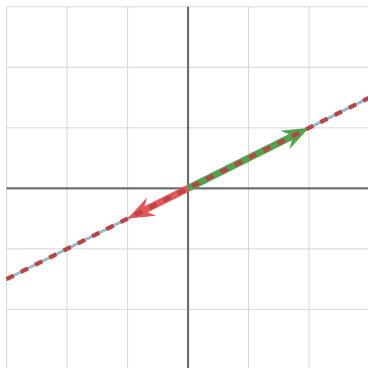
Shear: $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$



$\begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix}$: $\hat{i} \rightarrow [1, 2]$, $\hat{j} \rightarrow [3, 1]$



Dependent columns: $\begin{bmatrix} 2 & -1 \\ 1 & -0.5 \end{bmatrix}$



the whole plane is squished onto one line

green: where $\hat{i}$ lands (first column) red: where $\hat{j}$ lands (second column) grey: the grid before